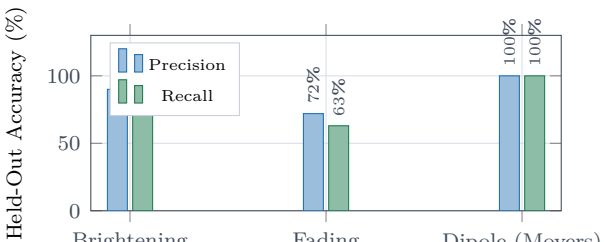
22 Features
SNR, χ^2, ellipticity, FWHM, mask frac

→ **HistGradient Boosting**
12,150 labels

→ **Isotonic Calib**
Calibrated P(real)
out-of-fold fit

Model Transparency (`spherexlens/ml/serve.py`):
- **Top-3 Reasons**: Human-readable feature effects (*"strong SNR +0.21, good PSF +0.12"*).
- **Decision Integrity**: Ranking aid only; **never overrides** deterministic triage.



Scientific Caveat from Model Card (`ml/MODEL_CARD.md`): The headline held-out score (Precision 0.96, PR-AUC 0.99) is inflated by dipoles because synthetic movers are dipoles while null-test artefacts almost never are. The honest measure of model skill is the per-kind table: brightening precision 0.90 / recall 0.82; fading 0.72 / 0.63. Public consensus votes (`/api/votes`) continuously provide real-world labels to retrain future models.